MRI Image Quality Metrics (IQMs)

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Abstract

Common MRI image quality metrics (IQMs) with their mathematical definitions and the fallback logic used in RadQy to avoid spurious NaNs.

Contents

1 Introduction

Metrics are reported per slice using foreground pixels **F** (from segmentation) and background pixels **B**. NaN/Inf inputs propagate as NaN. For BG-dependent metrics (SNR1, SNR2, SNR4, CNR, CJV, FBER), if background is missing or unusable (empty, zero variance, non-finite), we fall back to foreground homogeneity μ_F/σ_F.

Intensity and sign conventions. Prior to IQM calculation, slices are normalized to uint8 in the range $[0, 255]$; inputs outside this range are clipped. Foreground samples F are the *intensity values* inside the binary foreground mask (not the mask itself); background samples B are intensities where the mask is zero. IQM outputs are expected to be non-negative; any negative values indicate invalid inputs and are treated as QC failures.

2 Slice-Level IQMs

Grouped by descriptor family [1], [2], [3]. All are non-negative by construction; negative outputs indicate invalid inputs.

2.1 Spatial intensity / noise IQMs

2.1.1 Mean (MEAN)

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Algorithm 1: MEAN IQM

Input: Foreground pixels F (normalized to $[0, 255]$)
Output: μ_F or NaN
1. Remove non-finite entries from F
2. **if** F is empty **then**
3. **return** NaN
4. **end if**
5. $\mu_F \leftarrow \frac{1}{|F|}\sum_i F_i$
6. **return** μ_F

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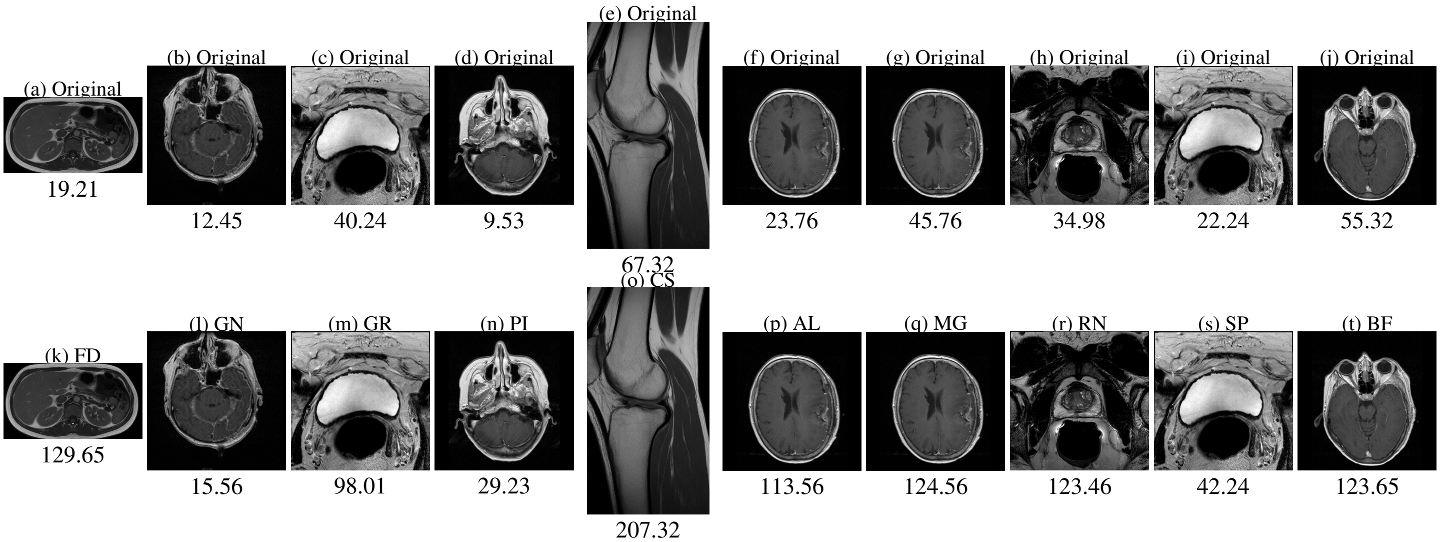

Figure 1: Mean values for sample MRI data. Top row (a–j): originals. Bottom row (k–t): corresponding artifacts (FD: field distortion, GN: Gaussian noise, GR: Gibbs ringing, PI: parallel imaging residuals, CS: chemical shift, AL: aliasing, MG: motion ghosting, RN: Rician noise, SP: spike/zipper, BF: bias field).

Foreground mean intensity:

$$\text{MEAN} = \mu_F = \frac{1}{N_F} \sum_i F_i$$

Non-negative with the $[0, 255]$ normalization; zero means black foreground. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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2.1.2 Range (RNG)

Foreground intensity range:

$$\mathrm{RNG} = \max(F) - \min(F)$$

Always non-negative.

Algorithm 2: RNG IQM

Input: Foreground pixels F
Output: RNG or NaN

1. Remove non-finite entries from F
2. **if** F is empty **then**
3.   **return** NaN
4. **end if**
5. RNG $\leftarrow \max(F) - \min(F)$
6. **return** RNG

2.1.3 Variance (VAR)

Foreground variance:

$$\mathrm{VAR} = \sigma_F^2 = \frac{1}{N_F}\sum_i (F_i - \mu_F)^2$$

Always non-negative.

Algorithm 3: VAR IQM

Input: Foreground pixels F
Output: σ_F^2 or NaN

1. Remove non-finite entries from F
2. **if** F is empty **then**
3.   **return** NaN
4. **end if**
5. $\mu_F \leftarrow \frac{1}{|F|}\sum_i F_i$
6. $\sigma_F^2 \leftarrow \frac{1}{|F|}\sum_i (F_i - \mu_F)^2$
7. **return** σ_F^2

2.1.4 Coefficient of Variation (CV)

Foreground coefficient of variation (percent):

$$\mathrm{CV} = \frac{\sigma_F}{\mu_F} \times 100$$

Non-negative; undefined (NaN) if $\mu_F = 0$.

Algorithm 4: CV IQM

Input: Foreground pixels F
Output: CV or NaN
Remove non-finite entries from F
if F is empty **then**
    return NaN
end if
$\mu_F \leftarrow \frac{1}{|F|} \sum_i F_i$
$\sigma_F \leftarrow \sqrt{\frac{1}{|F|} \sum_i (F_i - \mu_F)^2}$
if $\mu_F = 0$ **then**
    return NaN
end if
$\mathrm{CV} \leftarrow (\sigma_F / \mu_F) \times 100$
return CV

2.1.5 Contrast per Pixel (CPP)

Laplacian-filtered contrast, mean response of a 3×3 Laplacian:

$$f_1 = \frac{1}{8} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}, \quad \mathrm{CPP} = \mathrm{mean}\left(f_1 * F\right)$$

where $*$ is 2D convolution (foreground is median-filtered only when used for PSNR below). Can be positive or negative depending on dominant edge polarity; magnitude reflects local contrast.

2.1.6 Peak Signal-to-Noise Ratio (PSNR)

$$\mathrm{PSNR} = 10 \log_{10} \left(\frac{\max(F)^2}{\mathrm{MSE}(F, \hat{F})} \right)$$

where $\hat{F}$ is either the background **B** (if provided) or a median-filtered version of **F**. PSNR is typically positive; it can be low or negative if $\mathrm{MSE} \gg \max(F)^2$, indicating very poor fidelity.

Algorithm 5: PSNR IQM

Input: Foreground F, reference $\hat{F}$ (background or median-filtered F)
Output: PSNR or NaN
Remove non-finite entries from F and $\hat{F}$
if F or $\hat{F}$ is empty **then**
    return NaN
end if
$\mathrm{MSE} \leftarrow \frac{1}{|F|} \sum_i (F_i - \hat{F}_i)^2$
if $\mathrm{MSE} \leq 0$ **then**
    return NaN
end if
$\mathrm{PSNR} \leftarrow 10 \log_{10} \left(\frac{\max(F)^2}{\mathrm{MSE}} \right)$
return PSNR

2.1.7 SNR1

Foreground/background std; fallback to foreground stability if background unusable [4]:

$$\mathrm{SNR1} = \begin{cases} \dfrac{\sigma_F}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise (fg homogeneity)} \end{cases}$$

Non-negative by construction; NaN if denominators are zero or non-finite.

Algorithm 6: SNR1 IQM

Input: Foreground F, background B
Output: SNR1 or NaN

Remove non-finite entries from F, B
if F empty **then**
    return NaN
end if
$\mu_F \leftarrow \text{mean}(F)$; $\sigma_F \leftarrow \text{std}(F)$; $\sigma_B \leftarrow \text{std}(B)$
if σ_B finite and > 0 **then**
    return σ_F / σ_B
else
    return μ_F / σ_F
end if

2.1.8 SNR2

Foreground mean over background std; fallback to foreground stability if background unusable:

$$\text{SNR2} = \begin{cases} \dfrac{\mu_F}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

Non-negative by construction; NaN if denominators are zero or non-finite.

Algorithm 7: SNR2 IQM

Input: Foreground F, background B
Output: SNR2 or NaN

Remove non-finite entries from F, B
if F empty **then**
    return NaN
end if
$\mu_F \leftarrow \text{mean}(F)$; $\sigma_F \leftarrow \text{std}(F)$; $\sigma_B \leftarrow \text{std}(B)$
if σ_B finite and > 0 **then**
    return μ_F / σ_B
else
    return μ_F / σ_F
end if

2.1.9 SNR3

Foreground standard deviation divided by centered-foreground standard deviation (stability check):

$$\text{SNR3} = \frac{\sigma_F}{\sigma_{F-\mu_F}}$$

Non-negative; NaN if $\sigma_{F-\mu_F} = 0$ or non-finite.

Algorithm 8: SNR3 IQM

Input: Foreground F
Output: SNR3 or NaN

1. Remove non-finite entries from F
2. **if** F empty **then**
3. **return** NaN
4. **end if**
5. $\mu_F \leftarrow \text{mean}(F)$; $\sigma_F \leftarrow \text{std}(F)$; $\sigma_c \leftarrow \text{std}(F - \mu_F)$
6. **if** σ_c non-finite or $= 0$ **then**
7. **return** NaN
8. **end if**
9. **return** σ_F / σ_c

2.1.10 SNR4

Foreground mean over background mean; fallback to foreground stability if background unusable:

$$\text{SNR4} = \begin{cases} \dfrac{\mu_F}{\mu_B}, & \mu_B \text{ finite and } \mu_B \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

Non-negative by construction; NaN if denominators are zero or non-finite.

Algorithm 9: SNR4 IQM

Input: Foreground F, background B
Output: SNR4 or NaN

1. Remove non-finite entries from F, B
2. **if** F empty **then**
3. **return** NaN
4. **end if**
5. $\mu_F \leftarrow \text{mean}(F)$; $\mu_B \leftarrow \text{mean}(B)$; $\sigma_F \leftarrow \text{std}(F)$
6. **if** μ_B finite and $\neq 0$ **then**
7. **return** μ_F / μ_B
8. **else**
9. **return** μ_F / σ_F
10. **end if**

2.1.11 Contrast-to-Noise Ratio (CNR)

Foreground/background mean difference over background std; fallback to foreground stability if background unusable:

$$\text{CNR} = \begin{cases} \dfrac{\mu_F - \mu_B}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

Positive when foreground is brighter than background; negative when foreground is darker.

Algorithm 10: CNR IQM

Input: Foreground F, background B
Output: CNR or NaN

Remove non-finite entries from F and B
if F or B is empty **then**
    return NaN
end if
$\mu_F \leftarrow \text{mean}(F)$; $\mu_B \leftarrow \text{mean}(B)$
$\sigma_B \leftarrow \text{std}(B)$
if σ_B non-finite or $= 0$ **then**
    return $\mu_F/\text{std}(F)$
end if
$\text{CNR} \leftarrow (\mu_F - \mu_B)/\sigma_B$
return CNR

2.1.12 Patch Coefficient of Variation (CVP)

Foreground patch coefficient of variation:

$$\text{CVP} = \frac{\sigma_F}{\mu_F}$$

Non-negative; NaN if $\mu_F = 0$.

Algorithm 11: CVP IQM

Input: Foreground F
Output: CVP or NaN

Remove non-finite entries from F
if F empty **then**
    return NaN
end if
$\mu_F \leftarrow \text{mean}(F)$; $\sigma_F \leftarrow \text{std}(F)$
if $\mu_F = 0$ **then**
    return NaN
end if
return σ_F/μ_F

2.1.13 Coefficient of Joint Variation (CJV)

Joint variation between foreground and background; fallback to foreground stability if background unusable:

$$\text{CJV} = \begin{cases} \dfrac{\sigma_F + \sigma_B}{|\mu_F - \mu_B|}, & \text{finite and } |\mu_F - \mu_B| \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

Non-negative by construction; NaN if denominators are zero or non-finite.

Algorithm 12: CJV IQM

Input: Foreground F, background B
Output: CJV or NaN
Remove non-finite entries from F, B
if F empty **then**
    return NaN
end if
$\mu_F \leftarrow \text{mean}(F)$; $\mu_B \leftarrow \text{mean}(B)$
$\sigma_F \leftarrow \text{std}(F)$; $\sigma_B \leftarrow \text{std}(B)$
$d \leftarrow |\mu_F - \mu_B|$
if $d = 0$ or non-finite **then**
    return μ_F / σ_F
end if
return $(\sigma_F + \sigma_B)/d$

2.1.14 Entropy Focus Criterion (EFC)

Normalized Shannon entropy of foreground:

$$p_i = \frac{F_i}{\sqrt{\sum_j F_j^2}}, \quad \text{EFC} = -\frac{1}{\log N_F} \sum_i p_i \log p_i$$

Higher values suggest more ghosting/noise. Non-negative; NaN if $N_F \leq 1$ or norms are zero/non-finite.

Algorithm 13: EFC IQM

Input: Foreground F
Output: EFC or NaN
Remove non-finite entries from F
if F empty **then**
    return NaN
end if
$\text{norm} \leftarrow \sqrt{\sum_i F_i^2}$
if norm non-finite or $= 0$ **then**
    return NaN
end if
$p_i \leftarrow (F_i/\text{norm})$ clipped to $[10^{-12}, 1]$
$\text{EFC} \leftarrow -\frac{1}{\log |F|} \sum_i p_i \log p_i$
return EFC

2.1.15 Foreground-Background Energy Ratio (FBER)

Median energy ratio; fallback to foreground stability if background unusable:

$$\text{FBER} = \begin{cases} \dfrac{\text{median}(F^2)}{\text{median}(B^2)}, & \text{finite and median}(B^2) \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

Non-negative by construction; NaN if denominators are zero or non-finite.

Algorithm 14: FBER IQM

Input: Foreground F, background B
Output: FBER or NaN

1. Remove non-finite entries from F, B
2. **if** F empty **then**
3. **return** NaN
4. **end if**
5. $m_F \leftarrow \text{median}(F^2)$; $m_B \leftarrow \text{median}(B^2)$
6. **if** m_B finite and > 0 **then**
7. **return** m_F / m_B
8. **else**
9. **return** μ_F / σ_F
10. **end if**

2.1.16 Skewness (SKW)

Foreground skewness (third standardized moment):

$$\text{SKW} = \frac{\frac{1}{N_F} \sum_i (F_i - \mu_F)^3}{\sigma_F^3}$$

Can be negative (left-tailed), zero (symmetric), or positive (right-tailed).

Algorithm 15: SKW IQM

Input: Foreground F
Output: SKW or NaN

1. Remove non-finite entries from F
2. **if** F empty **then**
3. **return** NaN
4. **end if**
5. $\mu_F \leftarrow \text{mean}(F)$; $\sigma_F \leftarrow \text{std}(F)$
6. **if** $\sigma_F = 0$ or non-finite **then**
7. **return** NaN
8. **end if**
9. SKW $\leftarrow \frac{1}{|F|} \sum_i (F_i - \mu_F)^3 / \sigma_F^3$
10. **return** SKW

2.1.17 Excess Kurtosis (KURT)

Foreground excess kurtosis (fourth standardized moment minus 3):

$$\text{KURT} = \frac{\frac{1}{N_F} \sum_i (F_i - \mu_F)^4}{\sigma_F^4} - 3$$

Can be negative (platykurtic), zero (mesokurtic), or positive (leptokurtic); all non-NaN values are valid.

Algorithm 16: KURT IQM

Input: Foreground F
Output: KURT or NaN
1. Remove non-finite entries from F
2. **if** F empty **then**
3. **return** NaN
4. **end if**
5. $\mu_F \leftarrow \text{mean}(F)$; $\sigma_F \leftarrow \text{std}(F)$
6. **if** $\sigma_F = 0$ or non-finite **then**
7. **return** NaN
8. **end if**
9. $\text{KURT} \leftarrow \frac{1}{|F|} \sum_i (F_i - \mu_F)^4 / \sigma_F^4 - 3$
10. **return** KURT

2.2 Frequency-domain IQMs

Let $\mathcal{F}\{F\}(u, v)$ be the FFT magnitude of a slice. Masks Ω_L (central disk) and Ω_H (outer ring) define low/high bands [1].

2.2.1 Low-Frequency Response (LFR)

$$\text{LFR} = \frac{1}{|\Omega_L|} \sum_{(u,v)\in\Omega_L} |\mathcal{F}\{F\}(u, v)|$$

Always non-negative (FFT magnitude).

Algorithm 17: LFR IQM

Input: Foreground slice F, low-frequency mask Ω_L
Output: LFR or NaN
1. Remove non-finite entries from F
2. **if** F empty **then**
3. **return** NaN
4. **end if**
5. $M \leftarrow |\mathcal{F}\{F\}|$
6. **if** Ω_L empty **then**
7. **return** NaN
8. **end if**
9. $\text{LFR} \leftarrow \text{mean}(M \text{ over } \Omega_L)$
10. **return** LFR

2.2.2 High-Frequency Response (HFR)

$$\text{HFR} = \frac{1}{|\Omega_H|} \sum_{(u,v)\in\Omega_H} |\mathcal{F}\{F\}(u, v)|$$

Always non-negative (FFT magnitude).

Algorithm 18: HFR IQM

Input: Foreground slice F, high-frequency mask Ω_H
Output: HFR or NaN
1. Remove non-finite entries from F
2. **if** F empty **then**
3. **return** NaN
4. **end if**
5. $M \leftarrow |\mathcal{F}\{F\}|$
6. **if** Ω_H empty **then**
7. **return** NaN
8. **end if**
9. HFR $\leftarrow$ mean(M over Ω_H)
10. **return** HFR

2.2.3 Spectral SNR (SNRF)

$$\mathrm{SNRF} = \frac{\mathrm{HFR}}{\mathrm{LFR} + \epsilon}$$

with a small ϵ to avoid division by zero. Higher SNRF indicates sharper, higher-frequency content; lower SNRF indicates smoother slices. Always non-negative; undefined if bands are empty.

Algorithm 19: SNRF IQM

Input: Foreground slice F, masks Ω_L, Ω_H, ϵ
Output: SNRF or NaN
1. Compute LFR and HFR as above
2. **if** LFR or HFR is NaN **then**
3. **return** NaN
4. **end if**
5. **return** HFR/(LFR + ϵ)

2.3 Wavelet radiomics treated as IQMs

Using a 2D discrete wavelet transform yielding subbands W_k; each subsubsection refers to one subband (e.g., LH, HL, HH) [2].

2.3.1 Wavelet Coefficient Statistics (WCS)

$$\mathrm{WCS} = \frac{1}{|W_k|} \sum_{w \in W_k} |w|$$

Mean absolute coefficient magnitude; larger implies more detail energy. Always non-negative.

Algorithm 20: WCS IQM

Input: Slice F, wavelet subbands W_k (e.g., LH/HL/HH)
Output: WCS or NaN

1. Remove non-finite entries from F
2. **if** F empty **then**
3.   **return** NaN
4. **end if**
5. Compute one-level wavelet high-frequency subbands W_k
6. Collect coefficients $w \in W_k$; remove non-finite
7. **if** W_k empty **then**
8.   **return** NaN
9. **end if**
10. **return** $\frac{1}{|W_k|} \sum |w|$

2.3.2 Wavelet Coefficient Energy (WCE)

$$\mathrm{WCE} = \sum_{w \in W_k} w^2$$

Total energy in the subband; sensitive to noise and fine detail. Always non-negative.

Algorithm 21: WCE IQM

Input: Slice F, wavelet subbands W_k
Output: WCE or NaN

1. Remove non-finite entries from F; compute W_k as above
2. Collect finite coefficients w
3. **if** W_k empty **then**
4.   **return** NaN
5. **end if**
6. **return** $\sum w^2$

2.3.3 Wavelet Quadrant Sparsity (WQS)

$$\mathrm{WQS} = \frac{1}{|W_k|} \sum_{w \in W_k} \mathbf{1}\{|w| < \tau\}$$

Fraction of near-zero coefficients (threshold τ); higher means smoother, lower means textured. In $[0, 1]$ by definition.

Algorithm 22: WQS IQM

Input: Slice F, wavelet subbands W_k, threshold τ
Output: WQS or NaN
1. Remove non-finite entries from F; compute W_k as above
2. Collect finite coefficients w
3. **if** W_k empty **then**
4. **return** NaN
5. **end if**
6. **return** $\frac{1}{|W_k|} \sum \mathbf{1}\{|w| < \tau\}$

2.4 Texture radiomics IQMs (GLCM)

Let $P(i, j)$ be a normalized gray-level co-occurrence matrix (GLCM) [3]. Each metric is computed per direction and averaged.

2.4.1 Contrast

$$\text{Contrast} = \sum_{i,j} (i - j)^2 P(i, j)$$

Always non-negative.

Algorithm 23: GLCM Contrast IQM

Input: Quantized slice F; offsets for GLCM
Output: Contrast or NaN
1. Build normalized GLCM $P(i, j)$ over chosen offsets
2. **if** P empty **then**
3. **return** NaN
4. **end if**
5. **return** $\sum_{i,j} (i - j)^2 P(i, j)$

2.4.2 Dissimilarity

$$\text{Dissimilarity} = \sum_{i,j} |i - j| \, P(i, j)$$

Always non-negative.

Algorithm 24: GLCM Dissimilarity IQM

Input: GLCM $P(i, j)$
Output: Dissimilarity or NaN
1. **if** P empty **then**
2. **return** NaN
3. **end if**
4. **return** $\sum_{i,j} |i - j| \, P(i, j)$

2.4.3 Angular Second Moment (ASM)

$$\text{ASM} = \sum_{i,j} P(i, j)^2$$

Always non-negative.

Algorithm 25: GLCM ASM IQM

Input: GLCM $P(i, j)$
Output: ASM or NaN
1. **if** P empty **then**
2. **return** NaN
3. **end if**
4. **return** $\sum_{i,j} P(i, j)^2$

2.4.4 Energy

$$\text{Energy} = \sqrt{\text{ASM}}$$

Always non-negative.

Algorithm 26: GLCM Energy IQM

Input: GLCM $P(i, j)$
Output: Energy or NaN
1. Compute ASM as above
2. **if** ASM is NaN **then**
3. **return** NaN
4. **end if**
5. **return** $\sqrt{\text{ASM}}$

2.4.5 Homogeneity

$$\text{Homogeneity} = \sum_{i,j} \frac{P(i, j)}{1 + (i - j)^2}$$

Always non-negative; higher for smoother textures.

Algorithm 27: GLCM Homogeneity IQM

Input: GLCM $P(i, j)$
Output: Homogeneity or NaN
1. **if** P empty **then**
2. **return** NaN
3. **end if**
4. **return** $\sum_{i,j} \frac{P(i,j)}{1+(i-j)^2}$

2.4.6 Correlation

$$\text{Correlation} = \frac{\sum_{i,j}(i-\mu_i)(j-\mu_j)P(i,j)}{\sigma_i\sigma_j}$$

with $\mu_i, \mu_j, \sigma_i, \sigma_j$ the marginal means and standard deviations of P. May be negative if co-occurring gray levels are inversely related; NaN if marginals are degenerate.

Algorithm 28: GLCM Correlation IQM

Input: GLCM $P(i, j)$
Output: Correlation or NaN
1. **if** P empty **then**
2. **return** NaN
3. **end if**
4. Compute marginals $\mu_i, \mu_j, \sigma_i, \sigma_j$
5. **if** $\sigma_i = 0$ or $\sigma_j = 0$ or non-finite **then**
6. **return** NaN
7. **end if**
8. **return** $\frac{\sum_{i,j}(i-\mu_i)(j-\mu_j)P(i,j)}{\sigma_i\sigma_j}$

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3 Volume / Participant-Level IQMs

Slice-level IQMs are pooled per volume/participant for stability and domain-shift analysis [1], [4].

3.1 Slice-aggregate statistics

For any slice metric m_s over S slices:

$$m_{\text{mean}} = \frac{1}{S}\sum_{s=1}^{S} m_s, \quad m_{\text{median}} = \text{median}\{m_s\}, \quad m_{\text{std}} = \sqrt{\frac{1}{S}\sum_{s=1}^{S}(m_s - m_{\text{mean}})^2}.$$

Min/max and percentile aggregates are computed similarly to summarize outliers.

3.2 Failure Count (FAIL_FRAC)

Counts total non-finite IQM evaluations (NaN/Inf) across all slices/metrics for a participant. Higher counts flag poor FG/BG segmentation or corrupted slices:

$$\text{FAIL\_FRAC} = \sum_{s\in\text{slices}} \sum_{m\in\text{IQMs}} \mathbf{1}\{\text{IQM}_{m,s} \text{ is non-finite}\}.$$

References

[1] MRIQC: Advancing the automatic prediction of image quality in MRI from unseen sites. *PLOS ONE*, 2017.

[2] Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach. *Nature Communications*, 2014.

[3] Textural features for image classification. *IEEE Transactions on Systems, Man, and Cybernetics*, 1973.

[4] MRQy—An open-source tool for quality control of MR imaging data. *Medical physics*, 2020.